

M39 — Engage opens in-encounter modal, not a new tab

Phase: Phase 7 follow-on (post-M38, operator-feedback fix) **Status:** DONE **Blocked by:** M33 (Engage button on voice card), M35 (Engage deep-link route + instructor stations) **Blocks:** none **Estimated effort:** 0.5 day

1. Purpose

Operator feedback after M38:

"The system audio is working but it jumps out of the encounter window to a station window. It should stay in the encounter window during the conversation if the conversation is prompted from the encounter module."

M33 + M35 wired the Engage button as `<a target="_blank">`, which spawns a new browser tab. The instructor loses the encounter console context (telemetry, ECG, devices, scene injector) the moment they click into the chat. That's the wrong UX for an instructor playing a character — the console is exactly where they want to be when deciding what the character should say next.

M39 keeps the chat inside the encounter window via a modal `<dialog>` containing an `iframe` pointed at the existing `/portal/engage/{encounter_id}/{persona_id}` route. The `iframe` inherits cookies, follows the 303 to `/station/{join}/INST-{pid}`, and renders the same chat UI we already had — TTS / STT / heartbeat all intact (M37 audio reset + M38 key live-refresh still apply unchanged).

Closing the dialog blanks the `iframe src` so any in-flight audio inside it stops cleanly. A "■ Pop out" affordance inside the dialog keeps the old new-window behavior available for operators who want to drag the chat to a second monitor.

2. Structure

Files touched:

- `portal/templates/encounter_console.html` — appended a `<dialog`

`id="engage-dialog">` element at the end of the content block (outside `.console-grid`). The dialog body is an `<iframe id="engage-dialog-frame" src="about:blank" allow="microphone; autoplay">`. Header carries the modal title (set dynamically to `■ Engage · <persona name>`), a `■ Pop out` link, and a `"X Close"` button.

- `portal/static/encounter_console.js`:

- The engage element in the voice grid (rendered by

`bootVoices()`) changed from `<a target="_blank" href=...>` to `<button class="char-engage">` with `data-persona`, `data-persona-name`, and `data-engage-href` attributes. - New click handler binds each button to `openEngageDialog(btn)`. - New `openEngageDialog(btn)` reads the button's data attrs, sets the `iframe src` + dialog title + popout href, and calls `dialog.showModal()` (with a fallback to `setAttribute("open")` for browsers that don't support the native modal API). - `DOMContentLoaded` handler wires the close button + the native `close` event to blank the `iframe src`.

- `portal/static/encounter_console.css` — `.engage-dialog`,

`.engage-dialog::backdrop`, `.engage-dialog-header`, `.engage-dialog-actions`, `.engage-dialog-popout`, `.engage-dialog-close`, `.engage-dialog-frame` styles. Dialog is `min(900px, 92vw)` wide × `min(720px, 88vh)` tall, with a dark blue header bar (`#143b8a` matching the brand) and a translucent backdrop.

No backend change. The `/portal/engage/{eid}/{pid}` route is unchanged (still 303s to `/station/{join}/INST-{pid}`). The station chat template, JS, and TTS path are all unchanged.

No schema migration. No new dataclass field.

3. Uses

3.1 Instructor flow (in-encounter chat)

1. Instructor is on `/portal/room/encounter/{id}`. 2. Voice card lists every persona on the bed. 3. Click ■ **Engage** on a row. 4. `openEngageDialog(btn)` runs: - Reads `btn.dataset.engageHref` → `/portal/engage/{eid}/{persona_id}`. - Sets `iframe src` to that URL. - Sets dialog title to ■ Engage · `<persona display name>`. - Sets `popout href` to the same URL. - Calls `dialog.showModal()` — overlay appears centered on top of the console with a darkened backdrop. 5. Inside the `iframe`: the engage route 303s through to `/station/{join}/INST-{persona_id}`. Standard station chat UI loads (PTT button, mode toggle, chat log). Mic permission is inherited via `allow="microphone; autoplay"`. 6. Instructor presses PTT, speaks. STT → POST `/turn` → Haiku reply → TTS playback. M37 audio reset is in effect, M38 key live-refresh is in effect. Audio plays inside the `iframe`; the encounter console (telemetry, ECG, devices) remains visible behind the backdrop. 7. To return to the console: click X **Close** or press ESC. The `close` event handler sets `frame.src = "about:blank"` which tears down the in-flight `<audio>` element inside the `iframe` and the chat UI page itself — no zombie playback.

3.2 Pop-out path (preserved)

The "■ Pop out" link inside the dialog header is a plain `<a target="_blank">` pointing at the same engage URL. Operators who want to drag the chat to a second monitor still can — they just close the dialog and reopen the chat as a separate tab.

3.3 Multiple engages

Clicking Engage on a DIFFERENT persona while the dialog is open sets the `iframe` to the new persona's URL — same dialog instance, `src` swap. Setting `iframe src` on a live element triggers a full page reload inside the frame; the previous chat session's mic release + audio teardown happen naturally as the page unloads.

4. Functions (exported API surface)

Symbol	Where	Purpose
<code>openEngageDialog(btn)</code> (new)	<code>portal/static/encounter_console.js</code>	Read button data-attrs, set <code>iframe src</code> + dialog title + <code>popout href</code> , show the modal.
(close handlers)	same	Blank <code>iframe src</code> on <code>close</code> button click and on the native <code><dialog></code> <code>close</code> event (ESC key).
(rendered by <code>bootVoices</code>)	<code><button class="char-engage" data-*></code>	Per-persona engage triggers in the voice grid.
(markup)	<code><dialog id="engage-dialog"></code> in <code>encounter_console.html</code>	Modal container with <code>iframe</code> + header (title + <code>popout</code> + <code>close</code>).

5. Limitations

- **Iframe inherits the console's cookies + storage.** That's how

the engage route's `require_vault` dependency authenticates. Acceptable; same-origin iframe is the simplest path.

- ****Iframe height is fixed at 88vh — long chat logs scroll inside**

the iframe.** Acceptable; matches the station-page behavior. A future M40 could make the dialog vertically resizable.

- ****The native `<dialog>` API ships in modern browsers**

(Chrome 37+, Firefox 98+, Safari 15.4+).** The JS uses `showModal()` if available with a fallback to `setAttribute("open")` for older engines (which renders the dialog non-modally — still visible, just without the focus trap). The fallback is unlikely to be exercised in practice given the operator workstation profile.

- **Iframe `<audio>` cleanup relies on `src="about:blank"` on close.**

This works because navigating an iframe unloads the previous page, which fires the chat JS's pagehide listeners (and the browser tears down the `<audio>` element). If a future `station_chat.js` doesn't clean up on pagehide, audio could leak. Documented; not seen in practice today.

- ****The encounter console's polling continues while the dialog is**

open** (telemetry, transcript, state). Acceptable; the dialog doesn't pause the console, and a backdrop-click does not close the dialog (only the explicit Close button or ESC), so the instructor can't accidentally lose chat context.

- **No keyboard shortcut to open Engage.** Operators must click.

Out of scope; clicks are also what the iPad operator uses.

- **The dialog title shows only the persona name + display name.**

Could also show role/voice id, but visual noise wasn't worth it. Refine in a future M40 if operators ask.

6. Test status

Test file	Asserts	Status	Last run
tests/v7/test_engage_modal_dialog.py::test_encounter_console_includes_engage_dialog_markup	Console carries <code><dialog id="engage-dialog"> + iframe + close + popout + mic-allow</code> attrs	PASS	2026-05-27
...::test_engage_dialog_starts_with_blank_iframe_src	iframe initial <code>src="about:blank"</code> (no eager chat-session preload)	PASS	2026-05-27
...::test_engage_js_uses_button_and_opens_dialog_not_new_tab	Engage element is <code><button></code> (no <code>target="_blank"</code>), <code>openEngageDialog</code> function exists, click handler wired, <code>showModal + close + frame-blanking</code> present	PASS	2026-05-27

Test file	Asserts	Status	Last run
...::test_engage_dialog_popout_link_uses_engage_href	The "■ Pop out" anchor's href is set from <code>btn.dataset.engageHref</code> inside <code>openEngageDialog</code>	PASS	2026-05-27
...::test_engage_button_carries_required_data_attributes	Rendered button has <code>data-persona</code> , <code>data-persona-name</code> , <code>data-engage-href</code> so the click handler can wire the dialog	PASS	2026-05-27
Full v7 suite	255 passed, 1 skipped (M20 Playwright skip — unchanged)	PASS	2026-05-27

7. Change list

Date	Author	Change	Files
2026-05-27	claude-code	Initial M39 implementation: dialog markup + iframe; <code>bootVoices</code> renders engage as <code><button></code> ; <code>openEngageDialog</code> + close handlers; dialog CSS; 5 new tests	<code>portal/templates/encounter_console.html</code> , <code>portal/static/encounter_console.{js,css}</code> , <code>tests/v7/test_engage_modal_dialog.py</code> (new)

8. Open questions / known issues

- **Should backdrop-click close the dialog?** Today only explicit

Close button + ESC close. Backdrop-click would match common modal UX but risks losing chat state on accidental clicks. Stuck with safer default for now.

- **Should the dialog persist its position/size across opens?**

Probably not — operators rarely customize. The fixed centered placement is fine.

- **A "■ Minimize" toggle** (collapse the chat to a corner pill)

could be useful** for an operator who wants to inject scenes from the console without closing the chat. Out of scope for M39; a future M40 could add it.

- **The iframe currently re-loads the chat UI from scratch on each**

Engage click.** That means the chat history (stored in `enc.stations[INST-pid].history`) is server-side persistent — good, you don't lose context — but the visible chat-log resets to whatever the chat template renders on load. Operators may expect to "pick up where we left off"; verify in LAN test whether this needs additional state-restore on the chat UI.

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